Max Welz

Postdoctoral researcher in statistics – University of Zurich

max.welz@uzh.ch mwelz.github.io mwelz Google Scholar ORCID

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Research Interests

My research focuses on robust statistical inference and machine learning, particularly the analysis of categorical data (such as questionnaire responses) and inference on heterogeneous treatment effects. I develop methods that remain reliable under contamination, model misspecification, and distributional shifts, motivated by applications in economics, business, the social sciences, and health. I also develop and maintain open-source R packages.

Affiliations

University of Zurich **Zurich, CH**
Postdoctoral researcher in the group of Carolin Strobl *Since November 2024*
80% research, 20% statistical consulting and teaching

Erasmus University, Dept. of Econometrics and ERIM **Rotterdam, NL**
PhD Candidate *2020–2024*
Supervisors: Patrick Groenen and Andreas Alfons
Thesis title: *Robust categorical data analysis*
- Awarded the 2026 Dissertation Prize of the international *Psychometric Society*
- Runner-up for the 2025 *Van Zwet Award* of the *Netherlands Society for Statistics and Operations Research (VVSOR)*

Massachusetts Institute of Technology, Dept. of Economics **Cambridge, MA, USA**
Visiting PhD Student *Fall 2023*
Invited by Victor Chernozhukov and Mert Demirer

Erasmus Medical Center, Dept. of Public Health **Rotterdam, NL**
Researcher *2019–2025*
- Master's thesis on causal inference in lung cancer research
- Involvement in analyses of medical trials

Erasmus University, Dept. of Econometrics **Rotterdam, NL**
Research Assistant *2018–2019*
Research assistant to Philip Hans Franses (10 hours per week)

Education

Erasmus University **Rotterdam, NL**
PhD in Statistics *2025*
Date of defense: June 6, 2025

Erasmus University **Rotterdam, NL**
MSc in Econometrics *2020*

University of Mannheim **Mannheim, DE**
BSc in Economics (specialization in Econometrics) *2017*

Publications

Job market paper

Welz, M. (2026). Robust estimation and inference with categorical data [arXiv:2403.11954]. https://doi.org/10.48550/arXiv.2403.11954 *(under review at Journal of the American Statistical Association)*

This paper develops the theory of C-estimation, providing a general framework for contamination-robust estimation and inference with categorical data. Applications include the robust analysis of questionnaire responses. C-estimation was explicitly highlighted in the citations for my Psychometric Society Dissertation Prize and VVSOR Van Zwet Award runner-up recognition.

Articles in peer-reviewed journals

Welz, M. (2026). Robust estimation of polyserial correlation coefficients: A density power divergence approach. *Psychometrika*, *91*(2), 445–472. https://doi.org/10.1017/psy.2026.10091. [Preprint | GitHub]

Welz, M., Mair, P., & Alfons, A. (2026). Robust estimation of polychoric correlation. *Psychometrika*, *91*(1), 247–278. https://doi.org/10.1017/psy.2025.10066. [Preprint | GitHub]

Welz, M., van der Aalst, C., Alfons, A., Naghi, A., Heuvelmans, M., Groen, H. J., de Jong, P. A., Aerts, J., Oudkerk, M., de Koning, H., & ten Haaf, K. (2025). A comparative analysis of heterogeneity in lung cancer screening effectiveness in two randomised controlled trials. *Nature Communications*, *16, 8060*. https://doi.org/10.1038/s41467-025-63471-6

Alfons, A., & Welz, M. (2024). Open science perspectives on machine learning for the identification of careless responding: A new hope or phantom menace? *Social and Personality Psychology Compass*, *18*(2), e12941. https://doi.org/10.1111/spc3.12941. [Preprint | GitHub]

Franses, P. H., & Welz, M. (2022a). Evaluating heterogeneous forecasts for vintages of macroeconomic variables. *Journal of Forecasting*, *41*(4), 829–839. https://doi.org/10.1002/for.2835. [GitHub]

Franses, P. H., & Welz, M. (2022b). Forecasting real GDP growth for Africa. *Econometrics*, *10*(1). https://doi.org/10.3390/econometrics10010003. [GitHub]

Working papers

Bergh, D., Orlandi, I., Welz, M., Moelijker, R., & Tang, Y. (2026). Using matched-sample research designs to reduce non-random sampling hazards: Assessment and best practice recommendations. *(under review at Organizational Research Methods)*.

Welz, M., & Alfons, A. (2026). When respondents don't care anymore: Identifying the onset of careless responding [arXiv:2303.07167]. https://doi.org/10.48550/arXiv.2303.07167 *(under review at Behavior Research Methods)*

Welz, M., Archimbaud, A., & Alfons, A. (2026). Quantifying the impact of careless responding [PsyArXiv:8fj6p_v2]. https://doi.org/10.31234/osf.io/8fj6p_v2 *(under review at Psychometrika)*

Published open-source software

Welz, M., Alfons, A., Demirer, M., & Chernozhukov, V. (2026). *`GenericML`: Generic machine learning inference* [R package version 0.2.2]. https://CRAN.R-project.org/package=GenericML

Welz, M., Alfons, A., & Mair, P. (2026). *`robcat`: Robust categorical data analysis* [R package version 0.0.1]. https://CRAN.R-project.org/package=robcat

Welz, M., & Alfons, A. (2025). *carelessonset: Identifying the onset of careless responding* [R package version 0.0.1]. https://github.com/mwelz/carelessonset

Welz, M., ten Haaf, K., & Alfons, A. (2023). *predmod: Predictive modeling for heterogeneous treatment effects in medicine* [R package version 0.0.1]. https://github.com/mwelz/predmod

In progress

- Valid inference in distributionally misspecified models for questionnaire responses. *With Steffen Grønneberg (BI Norwegian Business School).*
- Debiased machine learning inference for extremal indices. *With Sebastian Engelke (University of Geneva).*
- When robustness and fast computation are for free: Stochastic approximation of high-dimensional ordinal models. *With Irini Moustaki (LSE), Andreas Alfons (Erasmus University), Giuseppe Alfonzetti (University of Udine).*
- Robust estimation of structural equation models for ordinal data. *With Patrick Mair (Harvard University) and Andreas Alfons (Erasmus University).*

Grants and Awards

Grants

- *Erasmus Trust Fund Grant* (€2,500), supporting my fall 2023 research visit to Massachusetts Institute of Technology.
- *Open Mind Call 2021 Health and Technology* (€19,995.70), Dutch Convergence Alliance. Joint grant with Andreas Alfons (Erasmus University) and Kevin ten Haaf (Erasmus MC) for the *CITRUS* project (acceptance rate: $\approx 18\%$).

Awards

- *2026 Dissertation Prize of the Psychometric Society*. Best PhD thesis in psychometrics and statistics for the social sciences; US$2,000 prize, IMPS 2026 keynote, and invited manuscript in *Psychometrika*.
- *Runner-up for the 2025 Van Zwet Award*. Annual award by the VVSOR for the best PhD thesis in statistics and operations research from a Dutch institution.
- *2024 Hannan Graduate Student Travel Award*, Institute of Mathematical Statistics. Support to attend JSM 2024, awarded for my research on robust categorical inference; one of ∼20 recipients worldwide.
- *Best PhD Student Presentation* at the 2022 annual meeting of the Dutch/Flemish Classification Society, Tilburg, Netherlands.
- *2021 Convergence Health and Technology Open Research Award* by the Dutch Convergence Alliance and open science initiatives. For commitment to open research; one of 12 recipients.

Teaching

FS = Fall semester, SS = Spring semester. Unless indicated otherwise, teaching was in English. Course evaluations are reported whenever available. Please click [here] for course syllabi and complete teaching evaluations.

University of Zurich (UZH; study programs in psychology)

- **2026–27**
  - *Statistics 1 (Part 1)* (FS, BSc, part of a module with a total of 18 ECTS, in German)
    This is an introductory statistics course attended by over 700 students. Role: Course coordinator. My

responsibilities include preparing and delivering 14 weekly lectures, co-designing and co-grading the final exam, as well as course administration. Evaluations are not yet available.

2025–26

- *Statistics 1 (Part 1)* (FS, BSc, part of a module with a total of 18 ECTS, in German)
  Role: Coordination of the tutorials and supervision of five TAs. My role was not evaluated.

- *Dirty Data: Outliers and Missing Values in Statistics and Psychology* (SS, 4 ECTS, seminar, MSc)
  This is a graduate seminar I designed myself. Annual attendance of about 30 students. My responsibilities included delivering 11 weekly lectures, designing and grading a final take-home exam, as well as course administration. Evaluation of my role: 5.8/6 (1 = "very dissatisfied", 6 = "very satisfied").

2024–25

- *Dirty Data: Outliers and Missing Values in Statistics and Psychology* (SS, 4 ECTS, seminar, MSc)
  Same role as in 2025–26. Evaluation of my role: 5.9/6 (1 = "very dissatisfied", 6 = "very satisfied").

Erasmus University Rotterdam (EUR; study programs in econometrics)

2023–24

- *Seminar in Machine Learning* (SS, 12 ECTS, BSc)
  In this undergraduate seminar, I supervised up to three groups of about four students each. Responsibilities included designing a topic, weekly feedback and meetings with each group, and grading of seminar reports. My groups did not submit evaluations.

2022–23

- *Seminar in Machine Learning* (SS, 12 ECTS, BSc)
  Same as in 2023–24. My groups did not submit evaluations.

- *Machine Learning* (FS, 4 ECTS; MSc)
  Role: Designing and grading midterm take-home assignments. About 200 students. My role was not evaluated.

2021–22

- *Seminar in Machine Learning* (SS, 12 ECTS, BSc)
  Same as in 2023–24. My groups did not submit evaluations.

2020–21

- *Academic Skills* (SS, 4 ECTS)
  Role: teaching assistant. Responsibilities: Teaching weekly tutorials of groups of about 30 students, grading of problem sets. Evaluation of my role: 4.4/5 (five-point Likert scale).

- *Seminar in Econometrics* (FS, 12 ECTS, BSc, online)
  In this undergraduate seminar, I supervised up to three groups of about four students each. Responsibilities included designing a topic, weekly feedback and meetings with each group, and grading of seminar reports. My groups did not submit evaluations.

2019–20

- *Academic Skills* (SS, 4 ECTS, BSc, online)
  Same role as in 2020–21.

- IT support for *Introduction to Statistics* (SS, 6 ECTS; BSc, in Dutch)
  Due to the onset of the pandemic, this course had to be moved online at short notice. I helped coordinate the course's transition to online delivery. My role was not evaluated.

2018–19

- *Introduction to Multivariate Statistics* (FS, 4 ECTS; BSc)
  Role: teaching assistant. Responsibilities: Teaching weekly tutorials of about 30 students. Evaluation of my role: 4.67/5 (five-point Likert scale).

Teaching qualification

- *University Delivery Qualification* at Erasmus University Rotterdam
  Development of essential didactic abilities such as motivating students, active learning, and presentation of information.

Supervision

Student theses supervised by me typically involve domain-specific applications of modern statistical methods such as machine learning and artificial intelligence.

- **University of Zurich; study programs in psychology:** I supervised two master's theses and four bachelor's theses.
- **Erasmus University Rotterdam; study programs in econometrics:** I supervised a total of 12 bachelor's theses as primary supervisor and 14 as second examiner, as well as three master's theses as primary supervisor and 14 as second examiner.

Community Service

Reviewing
Psychometrika (4), Behavior Research Methods (3), Journal of Statistical Software (1), Scandinavian Journal of Statistics (1)

Mandates
- Elected member of the Assembly of the Department of Psychology at the University of Zurich (2026–2028)
- Co-webmaster of the international Psychometric Society (since 2026)

Local Organization Committee
- *Data Science, Statistics & Visualization and European Conference on Data Analysis 2021* (joint conference). July 7–9, 2021. Erasmus University Rotterdam, NL (online)
- *Data Science, Statistics & Visualization 2020*. July 29–31, 2020. Duke University, USA (online)

Presentations

Invited Seminar Presentations
- 2027 (all scheduled): Charles University Prague, Toulouse Business School, University of Georgia
- 2026/10: University of Geneva, *Robust estimation and inference for categorical data*
- 2026/03: ETH Zurich, *Robust estimation and inference for categorical data (with an application to structural equation models)*
- 2026/02: Erasmus Medical Center Rotterdam, *Comparative analyses of heterogeneity in lung cancer screening effectiveness*
- 2025/10: London School of Economics, *Robust estimation and inference for categorical data (with an application to structural equation models)*

- 2025/02: Erasmus University Rotterdam, *Robust estimation and inference for categorical data*
- 2023/10: Massachusetts Institute of Technology, *Robust estimation and inference for categorical data*
- 2023/04: Tilburg University, *Identifying periods of careless responding in surveys: A deep learning approach*
- 2023/03: Erasmus University Rotterdam (PhD seminar), *On the impact of careless responding*
- 2022/06: Erasmus University Rotterdam (Social AI Seminar), *Identifying periods of careless responding in surveys: A deep learning approach*

Conference Presentations

*Invited talks are marked by an asterisk (*)*

2026

- 5th Workshop on Challenges in Categorical Data Analysis* (Athens, GR), *Robust estimation and inference for categorical data* (with discussion)
- Norwegian Psychometrics Gathering* (Oslo, NO), *Robust psychometrics*
- European Meeting of Statisticians (Lugano, CH), *Robust estimation and inference for categorical data*
- International Meeting of the Psychometric Society* (IMPS, Seoul, KR, keynote), *Robust psychometrics*
- International Workshop on Psychometric Computing (Psychoco, Padua, IT), *Robust pairwise estimation of factor analyses of categorical data*

2025

- IMPS (Minneapolis, USA), *Robust estimation of structural equation models*
- International Conference on Robust Statistics* (ICORS, Stresa, IT), *Robust estimation of reliability coefficients with rating-scale data*
- Deutsche Arbeitsgemeinschaft Statistik (Berlin, DE), *Robust estimation and inference for categorical data*
- Psychoco (Ghent, BE), *Robust estimation of structural equation models*

2024

- European Conference on Data Analysis* (Sopot, PL), *Robust estimation and inference in categorical data*
- 11th World Congress in Probability and Statistics (Bochum, DE), *Robust estimation and inference in categorical data*
- Joint Statistical Meetings (Portland, USA), *Robust estimation and inference in categorical data*
- ICORS (Washington DC, USA), *Robust estimation and inference in categorical data*
- Summer Meeting of the European Econometric Society (Rotterdam, NL), *Robust estimation and inference in categorical data*
- Netherlands Econometric Study Group (NESG, Maastricht, NL), *Robust estimation and inference in categorical data*
- Psychoco (Rotterdam, NL), *Detecting the onset of careless responding with the R package* `carelessonset`

2023

- CMStatistics* (Berlin, DE), *Robust parameter estimation in discrete data*
- ICORS (Toulouse, FR), *Robustness properties of correlation measures in ordinal discrete data*
- Psychoco (Zürich, CH), *A machine learning approach for identifying the onset of careless responding*
- Society for the Improvement of Psychological Science (SIPS, Padua, IT), *Balancing data privacy and reproducibility in open science: An introduction to differential privacy*

2022

- CMStatistics* (London, UK), *Identifying periods of careless responding in surveys: A deep learning approach*
- Annual Meeting of the Dutch-Flemish Classification Society (Tilburg, NL), *Identifying periods of careless responding in surveys: A deep learning approach*
- ICORS (Waterloo, CA), *Identifying periods of careless responding in surveys: a deep learning approach*
- SIPS (Victoria, CA), *Identifying periods of careless responding in surveys: A deep learning approach*
- useR! (online), *Generic machine learning inference on heterogeneous treatment effects using the package* `GenericML`

2021

- ICORS (Vienna, AT), *Outlier detection in rating-scale data via autoencoders*

Miscellaneous Presentations and Outreach

- Lecture to econometrics undergraduate students about econometrics in the health sciences. Invited by the econometrics study association at Erasmus University (FAECTOR). March 15, 2023.
- Various presentations in the reading group *"High-dimensional statistics"* at the Department of Econometrics at Erasmus University as a core member. Since February 2023.
- Various presentations in the reading group "Mathematical Statistics" at the math department of TU Delft. Since November 2021.

Scientific Network

Shorter research visits (<2 weeks)

- BI Norwegian Business School, Department of Data Science and Analytics (September 2026, invited by Steffen Grønneberg)
- London School of Economics, Department of Statistics (October 2025, invited by Irini Moustaki)
- Harvard University, Department of Psychology (July 2025, invited by Patrick Mair)

Membership in scientific organizations

- Psychometric Society
- Institute of Mathematical Statistics
- International Association for Statistical Computing
- Open Science Community Rotterdam
- Dutch-Flemish Classification Society

Media Coverage

- *Robuuste analyse van categorische gegevens: willekeurig antwoorden en verder* (EN: Robust analysis of categorical data: careless responding and beyond). STAtOR, forthcoming (fall 2026).
- *It is never too late to quit: lung cancer screening is particularly effective in former smokers*. Backbone Magazine, September 22, 2026.
- *Robust analysis of rating-scale data*. 'The Measure of it' blog (LSE), December 10, 2025.
- *Screening voor longkanker voorkomt vooral sterfgevallen bij ex-rokers* (EN: Lung cancer screening primarily prevents deaths among ex-smokers). nu.nl, November 24, 2025.
- *Longkankerscreening vooral effectief bij ex-rokers* (EN: Lung cancer screening is particularly effective in ex-smokers). Amazing Erasmus MC, November 24, 2025.
- *Breakthrough in personalised lung cancer screening*. Erasmus School of Economics newsletter, November 24, 2025.

Skills

Statistical computing and programming: R, Python, C++, C, Julia, MATLAB, Stata, Java, Bash, Zsh.

Tools and workflows: LaTeX, TensorFlow, Keras, PyTorch, git, agentic coding workflows using Codex and Claude Code.

Operating systems: Linux (e.g., Ubuntu, Fedora), macOS, Windows.

Languages: German (native), English (fluent), Dutch (fluent), French (intermediate).

Personal information

Citizen of Germany, Swiss residence permit B.

References

Andreas Alfons
Associate Professor of Statistics
Dept. of Econometrics
Erasmus University Rotterdam
alfons@ese.eur.nl

Irini Moustaki
Professor of Statistics
Dept. of Statistics
London School of Economics
i.moustaki@lse.ac.uk

Whitney K. Newey
Professor of Economics
Dept. of Economics
Massachusetts Institute of Technology
wnewey@mit.edu

Peter Bühlmann
Professor of Statistics
Dept. of Mathematics
ETH Zurich
buhlmann@stat.math.ethz.ch